

CS 6603: AI, Ethics, and Society

Final Exam

General information

The final exam is open-book, open-note, open-internet: you may consult any materials you want as long as you DO NOT interact with another human being. This means you may not post about the exam on Ed, text others about the exam, email others about the exam, talk on the phone about the exam, or otherwise gain support from another person by any means.

Do NOT copy and paste information from articles. If you copy and paste information, you will be flagged for plagiarism and will receive a 0 for the exam.

Exam Prompt (Proposal):

In this course, we have learned about the many abuses and misuses of AI from data privacy and ownership to algorithmic biases to qualifying and defining fairness. For the final exam, it is time to tie these course concepts to what we are seeing and experiencing with AI use in the news today.

Task 1 (25 pts): Find a Recent Public Artifact Addressing AI Misuse

Find a public artifact released within the last six months (February 2026 – July 2026) that identifies some aspects of AI misuse applied to a scenario and/or domain. The misuse must impact a regulated domain and/or a legally recognized protected class, as discussed in class. The public artifact must have some form of data evidence associated with it. Between the public artifact and data evidence, you should be able to identify at least 6 metrics demonstrating bias (see Task 3). Do not use the [dataset](#) we've examined in class.

Public artifacts include newspaper, magazine articles, books, blog posts, YouTube videos, podcasts, etc. Acceptable forms of data evidence include research publications, released datasets, results from surveys, etc.

In your report, provide the following information related to your artifact:

- Title
- Release date
- Link to artifact
- Application/scenario/domain of misuse
- Regulated domain/Protected class impacted
- Link to evidence

As an example (caveat: this public artifact was released beyond the six-month window):

- Public Artifact:
 - Title - "Can you make AI fairer than a judge? Play our courtroom algorithm game"
 - Released - October 17, 2019
 - Link - <https://www.technologyreview.com/s/613508/ai-fairer-than-judge-criminal-risk-assessment-algorithm/>
- Application/Scenario/Domain of Misuse: Criminal Risk Assessment (Predictive Algorithm)
- Regulated Domain/Protected Class: Education/Race
- Evidence: Dataset - <https://www.propublica.org/datastore/dataset/compas-recidivism-risk-score-data-and-analysis>

Note: The evidence/data does not need to have been released within the last six months, only the public artifact.

Task 2 (20 pts): Summary of Bias in Public Artifact

Provide a 1-2 paragraph summary of the bias that is identified by the public artifact. The summary should include references to definitions, concepts, and ideas discussed during this course (i.e., in lectures, assignments, cases studies, etc.). All references to definitions, concepts and ideas discussed in class must be highlighted in **bold** in your summary.

Task 3 (30 pts): Discussion of Artifact/Evidence Metrics Demonstrating Bias

Using the public artifact and associated data evidence, identify metrics demonstrating bias in at least 6 of the 8 categories below:

- Privileged/unprivileged groups
- Misleading graphs (Explain why a graph shown is misleading)
- Source(s) of data bias
- Source(s) of sampling bias
- Sampling methods used to collect data
- Correlations found in the data
- Outcome measures: averages, standard deviations, quartiles, frequency distributions, margins of error
- Bias & Fairness (or other) metrics used to identify differences in outcomes

Provide an explanation for each metric, including its significance, relevance to the category, and any quantifiable values associated with them as applicable. There should be detailed analysis for each category explaining the significance of these metrics. Screenshots and copy- pasted graphics/evidence from the artifact alone will not be accepted. There should be enough details provided for us to validate your ability to synthesize course concepts based on the overarching topics.

Note: Each bullet is considered one category. Listing the averages and standard deviations would count as one category. You must select at least six distinct categories to obtain full credit.

Task 4 (25 pts): Propose a Method for Mitigating Bias

Within your report, pinpoint an issue associated with one of the measurable metrics outlined above (Task 3) that, if resolved, could potentially alleviate bias and/or unfairness.

Develop a strategy to tackle the identified issue, ensuring it aligns with concepts covered in the lectures. Provide a detailed explanation of the method using pseudo-code or a Python script with comments.

For the proposed bias mitigation method, specify the data inputs and outputs. Additionally, analyze any expected alterations in outcomes resulting from the implementation of the method.

Note: Working code or quantitative results are not required.

Submission:

The final exam should be submitted as a PDF in JDF format.

The file name for submission is GTuserName_Final_Exam.pdf. Reports that are not neat and well organized will receive up to a 10-point deduction.